



Chapter 1- Introduction

Topics covered



- ✧ FAQs about software Engineering.
- ✧ Professional and ethical responsibility

What is Software?



- ✧ software is not just the programs but also all associated **documentation** and **configuration** data that is needed to make these programs operate correctly.
- ✧ Software engineers are concerned with developing software products, i.e., software which can be sold to a customer.

Software products



✧ Generic products

- Stand-alone systems that are marketed and sold to any customer who wishes to buy them.
- Examples – PC software such as graphics programs, project management tools; CAD software; software for specific markets such as appointments systems for dentists.

✧ Customized products

- Software that is commissioned by a specific customer to meet their own needs.
- Examples – embedded control systems, air traffic control software, traffic monitoring systems.

Frequently asked questions about software engineering



Question	Answer
What is software?	Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
What are the attributes of good software?	Good software should deliver the required functionality and performance to the user and should be maintainable, dependable and usable.
What is software engineering?	Software engineering is an engineering discipline that is concerned with all aspects of software production.
What are the fundamental software engineering activities?	Software specification, software development, software validation and software evolution.
What is the difference between software engineering and computer science?	Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
What is the difference between software engineering and system engineering?	System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this more general process.

Frequently asked questions about software engineering



Question	Answer
What are the key challenges facing software engineering?	Coping with increasing diversity, demands for reduced delivery times and developing trustworthy software.
What are the costs of software engineering?	Roughly 60% of software costs are development costs, 40% are testing costs. For custom software, evolution costs often exceed development costs.
What are the best software engineering techniques and methods?	While all software projects have to be professionally managed and developed, different techniques are appropriate for different types of system. For example, games should always be developed using a series of prototypes whereas safety critical control systems require a complete and analyzable specification to be developed. You can't, therefore, say that one method is better than another.
What differences has the web made to software engineering?	The web has led to the availability of software services and the possibility of developing highly distributed service-based systems. Web-based systems development has led to important advances in programming languages and software reuse.

Software engineering



- ✧ Software engineering is an **engineering discipline** that is concerned with **all aspects of software production** from the early stages of system specification through to maintaining the system after it has gone into use.
- ✧ Engineering discipline
 - Using appropriate **theories and methods** to solve problems bearing in mind organizational and financial constraints.
- ✧ All aspects of software production
 - Not just technical process of development. Also project management and the development of tools, methods etc. to support software production.

Essential attributes of good software



Product characteristic	Description
Maintainability	Software should be written in such a way so that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.
Dependability and security	Software dependability includes a range of characteristics including reliability, security and safety. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.
Efficiency	Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable and compatible with other systems that they use.

What's the difference between software engineering and computer science?



- ✧ Computer science is concerned with the **theories and methods** that underlie computers and software systems.
- ✧ Software engineering is concerned with the **practical problems** of producing software.

What is the difference between software engineering and system engineering?



- ✧ System engineering is concerned with **hardware development, policy and process design and system deployment.**
- ✧ System engineers are involved in specifying the system, defining its overall architecture and then integrating the different parts to create the finished system.

Importance of software engineering



- ✧ Individuals and society rely on advanced software systems.
- ✧ Able to produce reliable(failure free)and trustworthy systems economically and quickly.

Software process activities



Software process is the **set of activities** and associated results that produce a software product.

- ✧ **Software specification**, where customers and engineers define the software that is to be produced and the constraints on its operation.
- ✧ **Software development**, where the software is designed and programmed.
- ✧ **Software validation**, where the software is checked to ensure that it is what the customer requires.
- ✧ **Software evolution**, where the software is modified to reflect changing customer and market requirements.

What is a software process model?

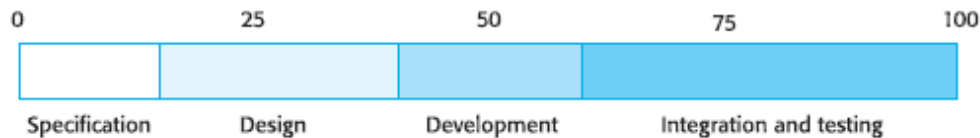


- ❖ *The waterfall approach* :process phases such as requirements specification, software design, implementation, testing and so on. After each stage is defined it is ‘signed-off’, and development goes on to the following stage.
- ❖ *Iterative development* :activities of specification, development and validation. An initial system is rapidly developed from very abstract specifications. This is then refined with customer input to produce a system that satisfies the customer’s needs.
- ❖ *Component-based software engineering (CBSE)* :technique assumes that parts of the system already exist. The system development process focuses on integrating these parts rather than developing them from scratch.

What are the costs of software engineering?



Waterfall model



Iterative development



Component-based software engineering



Development and evolution costs for long-lifetime



What are the key challenges facing software engineering?



✧ Heterogeneity

- Increasingly, systems are required to operate as distributed systems across networks that include different types of computer and mobile devices.

✧ The delivery challenge

- Many traditional software engineering techniques are time-consuming. The time they take is required to achieve software quality.

What are the key challenges facing software engineering?



✧ Security and trust

- As software is intertwined with all aspects of our lives, it is essential that we can trust that software.

✧ Scale

- Software has to be developed across a very wide range of scales, from very small embedded systems in portable or wearable devices through to Internet-scale, cloud-based systems that serve a global community.

Software engineering ethics



- ✧ Software engineering involves wider responsibilities than simply the application of technical skills.
- ✧ Software engineers must behave in an **honest** and **ethically responsible** way if they are to be respected as professionals.
- ✧ Ethical behaviour is more than simply upholding the law but involves following a set of principles that are morally correct.

Professional and Ethical responsibility



✧ Confidentiality

- Engineers should normally respect the confidentiality of their employers or clients irrespective of whether or not a formal confidentiality agreement has been signed.

✧ Competence

- Engineers should not misrepresent their level of competence. They should not knowingly accept work which is outwit their competence.

Professional and Ethical responsibility



✧ Intellectual property rights

- Engineers should be aware of local laws governing the use of intellectual property such as patents, copyright, etc. They should be careful to ensure that the intellectual property of employers and clients is protected.

✧ Computer misuse

- Software engineers should not use their technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses).

ACM/IEEE Code of Ethics



- ✧ Professional societies and institutions have an important role to play in setting ethical standards. Organizations such as the ACM (Association for Computing Machinery), the IEEE (Institute of Electrical and Electronic Engineers) .
- ✧ Members of these organizations undertake to follow that code when they sign up for membership.
- ✧ The ACM and the IEEE have cooperated to produce a joint code of ethics and professional practice.

Rationale for the code of ethics



- *Computers have a central and growing role in commerce, industry, government, medicine, education, entertainment and society at large.* Software engineers are those who contribute by direct participation or by teaching, to the analysis, specification, design, development, certification, maintenance and testing of software systems.
- *Because of their roles in developing software systems, software engineers have significant opportunities to do good or cause harm, to enable others to do good or cause harm, or to influence others to do good or cause harm. To ensure, as much as possible, that their efforts will be used for good, software engineers must commit themselves to making software engineering a beneficial and respected profession.*

The ACM/IEEE Code of Ethics



Software Engineering Code of Ethics and Professional Practice

ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices

PREAMBLE

The short version of the code summarizes aspirations at a high level of the abstraction; the clauses that are included in the full version give examples and details of how these aspirations change the way we act as software engineering professionals. Without the aspirations, the details can become legalistic and tedious; without the details, the aspirations can become high sounding but empty; together, the aspirations and the details form a cohesive code.

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following Eight Principles:

Ethical principles



1. PUBLIC - Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.
5. MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues.
8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Ethical dilemmas



- ✧ Disagreement in principle with the policies of senior management.
- ✧ Your employer acts in an unethical way and releases a safety-critical system without finishing the testing of the system.
- ✧ Participation in the development of military weapons systems or nuclear systems.